Experiment No. 14 – Perspective Transformation

1. Aim

To perform Perspective Transformation on a given image using Python and OpenCV.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imread(filename)

if image is None:
    print("Image loading failed!")
else:
    # Convert BGR to RGB
    image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Get image dimensions
    rows, cols = image.shape[:2]
```

```
# Select four points from original image
pts1 = np.float32([
    [0.15*cols, 0.15*rows],
    [0.85*cols, 0.10*rows],
    [0.95*cols, 0.85*rows],
    [0.05*cols, 0.90*rows]
])

# Destination points
pts2 = np.float32([
    [0, 0],
    [cols, 0],
    [cols, rows],
    [0, rows]
])

# Calculate transformation matrix
matrix = cv2.getPerspectiveTransform(pts1, pts2)

# Apply perspective transformation
transformed = cv2.warpPerspective(
    image, matrix, (cols, rows)
)

# Display results
plt.figure(figsize=(12, 6))
```

```
plt.subplot(1, 2, 1)
plt.imshow(image)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(transformed)
plt.title("Perspective Transformed Image")
plt.axis("off")

plt.show()
```

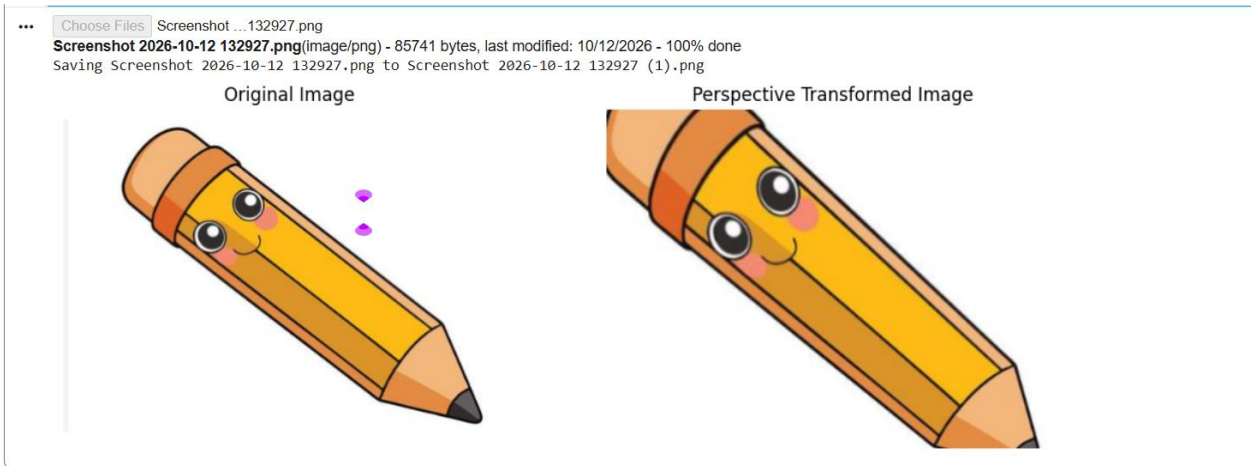
3. Expected Output

The output should display two images:

- Original Image
- Perspective Transformed Image

The transformed image will show the effect of changing the perspective of the selected four corner points.

4. Output



The original image and perspective-transformed image are displayed successfully.

5. Result

Thus, Perspective Transformation was successfully performed on the given image using Python and OpenCV.